

- Determinant Soru Çözüm -

1) $\det(A) = 0$ yapan tüm x değerlerini bulunuz

a) $A = \begin{pmatrix} x-3 & 3 \\ 2 & x+2 \end{pmatrix}_{2 \times 2}$

b) $A = \begin{pmatrix} x-3 & 0 & 0 \\ 0 & x & 1 \\ 0 & 2 & x-1 \end{pmatrix}_{3 \times 3}$

Gözüm:

a) $\det(A) = (x-3) \cdot (x+2) - 3 \cdot 2$

$$= x^2 - x - 6 - 6$$

$$= x^2 - x - 12$$

$$\Rightarrow (x-4)(x+3) = 0 \Rightarrow \boxed{x=4} \text{ veya } \boxed{x=-3}$$

b) $\det(A) = (x-3) \cdot (-1)^{1+1} \begin{vmatrix} x & 1 \\ 2 & x-1 \end{vmatrix} + 0 \nearrow + 0 \nearrow$

$$= (x-3) (x \cdot (x-1) - 2)$$

$$= (x-3) \cdot (x^2 - x - 2)$$

$$= (x-3) (x-2) (x+1)$$

$$\Rightarrow \det(A) = 0 \Leftrightarrow \boxed{x=3} \text{ veya } \boxed{x=2} \text{ veya } \boxed{x=-1}$$

2) $\begin{vmatrix} \sin x & \cos x & 1 \\ \cos x & -\sin x & 0 \\ -\sin x & -\cos x & 0 \end{vmatrix}$ determinantının x 'ten bağımsız olduğunu gösteriniz.

Çözüm :

$$\det(A) = 1 \cdot (-1)^{1+3} \cdot \begin{vmatrix} \cos x & -\sin x \\ -\sin x & -\cos x \end{vmatrix} + 0 \cdot \dots + 0 \cdot \dots$$

$$= 1 \cdot (-1) \cdot (-\cos^2 x - \sin^2 x)$$

$$= -1 \cdot \underbrace{(\cos^2 x + \sin^2 x)}_1 = -1$$

olup sonuç x 'ten bağımsızdır!

3) $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ olmak üzere $\det(A) = \frac{1}{2} \begin{vmatrix} \text{tr}(A) & \text{tr}(A^2) \\ 1 & \text{tr}(A) \end{vmatrix}$ olduğunu gösteriniz.

Çözüm:

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \quad \text{ve} \quad A^2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{pmatrix}$$

$$\frac{1}{2} \begin{vmatrix} \text{tr}(A) & \text{tr}(A^2) \\ 1 & \text{tr}(A) \end{vmatrix} = \frac{1}{2} \begin{vmatrix} a+d & a^2+2bc+d^2 \\ 1 & a+d \end{vmatrix}$$

$$= \frac{1}{2} \left((a+d)(a+d) - 1 \cdot (a^2+2bc+d^2) \right)$$

$$= \frac{1}{2} \cdot (\cancel{a^2} + 2ad + \cancel{d^2} - \cancel{a^2} - 2bc - \cancel{d^2})$$

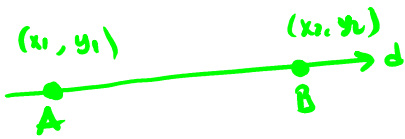
$$= ad - bc$$

$$= \det(A)$$

4) (x_1, y_1) ve (x_2, y_2) noktalarından geçen doğrunun denkleminin

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0 \quad \text{şeklinde yazılabileceğini gösteriniz.}$$

Görüm:



$$d: \quad y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} \cdot (x - x_1)$$

$$y - y_2 = \frac{y_2 - y_1}{x_2 - x_1} \cdot (x - x_2)$$

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 1 \cdot (-1)^{1+3} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + 1 \cdot (-1)^{2+3} \begin{vmatrix} x & y \\ x_2 & y_2 \end{vmatrix} + 1 \cdot (-1)^{3+3} \begin{vmatrix} x & y \\ x_1 & y_1 \end{vmatrix}$$

$$= x_1 \cdot y_2 - \underline{\underline{y_1 \cdot x_2}} - x \cdot y_2 + \underline{\underline{y \cdot x_2}} + x \cdot y_1 - \underline{\underline{y \cdot x_1}} + \underline{\underline{x_1 \cdot y_1}} - \underline{\underline{x_1 \cdot y_1}}$$

$$\Rightarrow x_1 \cdot y_2 - y_1 \cdot x_2 - x y_2 + y x_2 + x y_1 - y x_1 + x_1 \cdot y_1 - x_1 y_1 = 0$$

$$\Rightarrow x_1 \cdot y_2 - x \cdot y_2 + x y_1 - x_1 \cdot y_1 = y_1 \cdot x_2 - y \cdot x_2 + y x_1 - x_1 \cdot y_1$$

$$\Rightarrow y_2(x_1 - x) - y_1(x_1 - x) = y_1(x_2 - x_1) - y(x_2 - x_1)$$

$$\Rightarrow (x_1 - x)(y_2 - y_1) = (x_2 - x_1)(y_1 - y)$$

$$\Rightarrow (x_2 - x_1)(y - y_1) = (x - x_1)(y_2 - y_1)$$

$$\Rightarrow \boxed{y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} \cdot (x - x_1)}$$

5) Ügensel (alt, üst veya köşegen) matrislerin determinantları esas köşegen elemanlarının çarpımıdır. Örneğin

$$\begin{vmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix} = a_{11} \cdot (-1)^{1+1} \begin{vmatrix} a_{22} & 0 & 0 \\ a_{32} & a_{33} & 0 \\ a_{42} & a_{43} & a_{44} \end{vmatrix} \\
 = a_{11} \cdot \underbrace{(-1)^{1+1}}_{+1} \cdot a_{22} \cdot \underbrace{(-1)^{1+1}}_{+1} \cdot \begin{vmatrix} a_{33} & 0 \\ a_{43} & a_{44} \end{vmatrix} \\
 = a_{11} \cdot a_{22} \cdot a_{33} \cdot a_{44}$$

Buna göre

a) $A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 18 & 3 & 0 & 0 \\ 35 & 41 & 5 & 0 \\ 120 & 18 & 12 & 1/3 \end{pmatrix} \Rightarrow \det(A) = 2 \cdot 3 \cdot 5 \cdot \frac{1}{3} = 10$

b) $B = \begin{pmatrix} 100 & 200 & 300 & 400 \\ 0 & 0 & 500 & 600 \\ 0 & 0 & 700 & 800 \\ 0 & 0 & 0 & 900 \end{pmatrix} \Rightarrow \det(A) = 0$